Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- **Leave-One-Out Cross-Validation (LOOCV)**
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- Convex Optimization
  Reading • 8 min
- Linear Programming
  Reading • 15 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Leave-One-Out Cross-Validation (LOOCV)

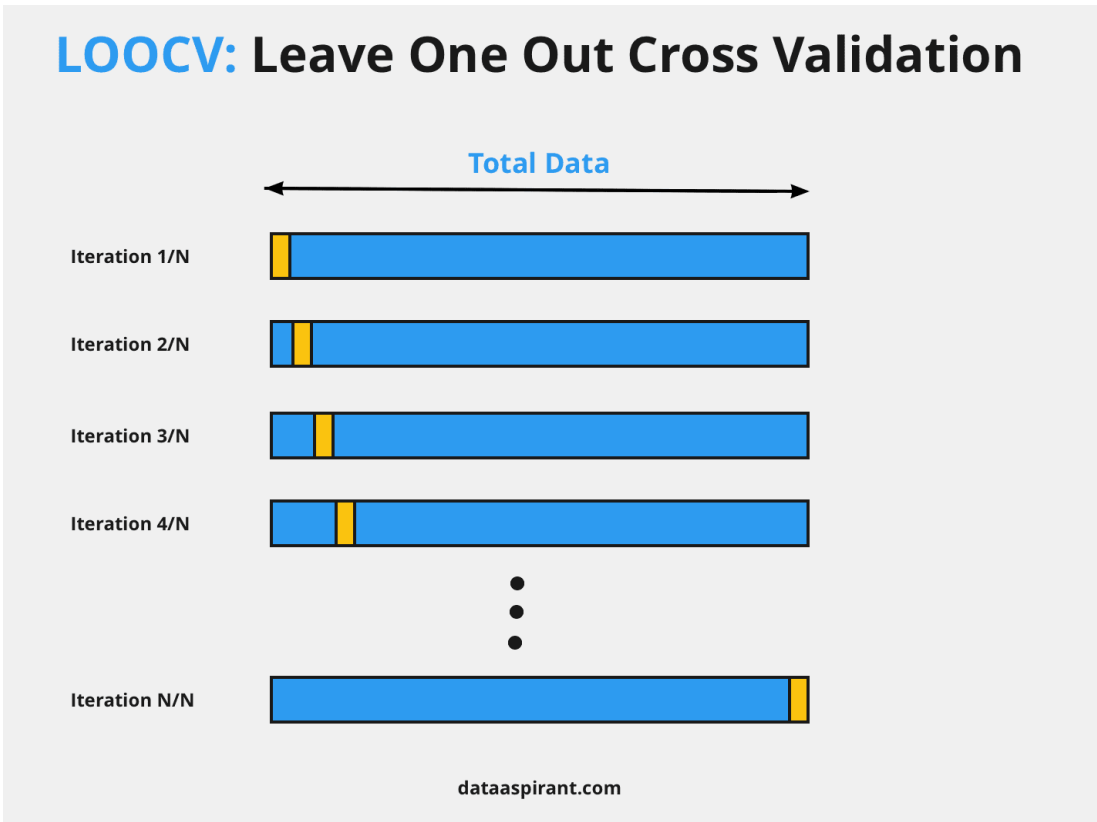
0:10 / 2:17 2x

Leave-One-Out Cross-Validation (LOOCV) is a special case of K-Fold Cross-Validation where K equals the number of data points in the dataset. In LOOCV, each data point is used once as the validation set, while the rest serve as the training set. This process is repeated for each data point, and the results are averaged. LOOCV is very thorough but can be computationally expensive, especially for large datasets, since the model is trained as many times as there are data points.

Steps:

1. **Select a Data Point:** For each iteration, select one data point to be the validation set.
2. **Train the Model:** Train the model on the remaining N-1 data points.
3. **Validate the Model:** Test the model on the single data point left out.
4. **Repeat:** Repeat this process for each data point in the dataset so that every data point is used as a validation set exactly once.
5. **Average the Results:** Calculate the average performance metric across all N iterations to estimate the model's generalization ability.

Cross Validation - Data Aspirant (20 minutes)



Conclusion

Cross-validation is a powerful and essential technique in the toolkit of any data scientist or statistician, as it provides a robust way to evaluate a model's ability to generalize to unseen data. By systematically partitioning the data into training and validation sets, cross-validation helps to mitigate the risk of overfitting, ensuring that the model is not just memorizing the training data but truly learning the underlying patterns.

K-Fold Cross-Validation, with its balance of efficiency and reliability, is particularly useful in practice, providing a solid estimate of model performance by averaging results across multiple folds. On the other hand, Leave-One-Out Cross-Validation (LOOCV) offers a thorough evaluation, especially for smaller datasets, although it comes with a higher computational cost.

Ultimately, cross-validation enables more informed decisions in model selection and tuning, leading to models that perform well not just on the data we have, but on the data we have yet to see. As you move forward in your work, understanding and applying these techniques will be crucial in building predictive models that are both accurate and robust.

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →